The Metaplectic Lift of Quintic Conifold Monodromy: Genus-Two and Genus-Three Crossing Blocks

A computational research note

16 October 2026

Abstract

The conifold monodromy of the mirror quintic, expressed in Movasati's symplectic period basis, is a rank-one shear. Its metaplectic lift is therefore a one-directional Gaussian operator on the BCOV wave function. We derive the exact connected-graph corrections through genus three, with all rational symmetry factors; give a transport law for genus-two and genus-three ambiguity variations; and identify an intrinsic family of polarization-covariant completed insertions. The calculation gives an exact local monodromy trivialization in the vanishing-cycle polarization and supports crossing-adapted *completed* blocks, but does not by itself eliminate the holomorphic ambiguity. Actual cancellation of the previously computed Yamaguchi–Yau generator cocycles requires pulling back the homogeneous amplitudes and the genus-one half-form normalization; that separate quintic-specific comparison is not claimed here.

1 Conifold shear and the precise meaning of the lift

Normalize the mirror coordinate by $s = 5^5 z$ and use the period ordering of Movasati, $\Pi = (x_1, x_2, x_3, x_4)^T$, with symplectic pairing $\langle x_1, x_3\rangle = \langle x_2, x_4\rangle = 1$. The conifold matrix is

$$M_c = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad (x_2, x_4) \longmapsto (x_2 + x_4, x_4). \tag{1}$$

It is a rank-one unipotent symplectic shear, not a mere substitution $s \mapsto 1 - s$. The wave function is defined on the *two-dimensional* space of homogeneous A-periods; for this shear only the x_2 direction participates. With $x = x_2$, $p = \partial_x \mathcal{F}_0$ at fixed x_1 and $h = \lambda^2$, we adopt the following heat-operator convention:

$$\boxed{Z_\beta(y; h) = \exp\left(\frac{\beta h}{2}\partial_y^2\right) Z_0(y; h), \qquad Z_0(x; h) = \exp \sum_{g\geq 0} h^{g-1}\mathcal{F}_g(x).} \tag{2}$$

Its Gaussian kernel is formally $\exp[-(y-x)^2/(2\beta h)]$. Its saddle equation reads $y = x - \beta\mathcal{F}_0'(x)$; hence the sign choice for the shear $y = x + p$ of (1) is $\beta = -1$. The integral is understood as a formal steepest-descent expansion. The inverse representation, a different phase convention for p, or the opposite monodromy orientation reverses this sign, not the structure of the resulting formulas. Global metaplectic phases and the half-form normalization must be tracked separately.

Let $\chi = \chi(y)$ solve the saddle equation and introduce

$$A = 1 - \beta \mathcal{F}_0''(\chi), \qquad \boxed{\kappa = \frac{\beta}{1 - \beta \mathcal{F}_0''(\chi)}}. \tag{3}$$

The classical and one-loop actions are

$$\mathcal{F}_0^{(\beta)}(y) = \mathcal{F}_0(\chi) - \frac{(y-\chi)^2}{2\beta}, \tag{4}$$

$$\mathcal{F}_1^{(\beta)}(y) = \mathcal{F}_1(\chi) - \frac{1}{2}\log A, \tag{5}$$

up to a y-independent metaplectic normalization. All derivatives below are ∂_x at fixed x_1 and are evaluated at $x = \chi(y)$.

2 Exact genus-two correction

Set $f_{gn} = \partial_x^n \mathcal{F}_g(\chi)$, with $f_{g0} = \mathcal{F}_g(\chi)$. Wick expansion of the Gaussian to order h^1 yields

$$\boxed{\begin{aligned} &\mathcal{F}_2^{(\beta)} = f_{20} + \Gamma_2, \\ &\quad \Gamma_2 = \frac{\kappa}{2}(f_{12} + f_{11}^2) + \kappa^2 \left(\frac{f_{04}}{8} + \frac{f_{03} f_{11}}{2} \right) + \frac{5\kappa^3}{24} f_{03}^2. \end{aligned}} \tag{6}$$

The coefficients $1/2, 1/8, 5/24$ arise from exact Wick-pairing counts, not from fitting quintic coefficients. In particular, the bare $\mathcal{F}_2$ is *not* invariant: it is accompanied by a uniquely determined connected lower-genus Gaussian correction.

3 Exact genus-three correction

The full h^2 calculation gives $\mathcal{F}_3^{(\beta)} = f_{30} + \Gamma_3$, where

$$\begin{aligned} \Gamma_3 = {} & \kappa \left(f_{11} f_{21} + \frac{1}{2} f_{22} \right) \\ & + \kappa^2 \left(\frac{1}{2} f_{03} f_{21} + \frac{1}{2} f_{11}^2 f_{12} + \frac{1}{2} f_{11} f_{13} + \frac{1}{4} f_{12}^2 + \frac{1}{8} f_{14} \right) \\ & + \kappa^3 \left(\frac{1}{6} f_{03} f_{11}^3 + f_{03} f_{11} f_{12} + \frac{5}{12} f_{03} f_{13} + \frac{1}{4} f_{04} f_{11}^2 + \frac{1}{4} f_{04} f_{12} + \frac{1}{8} f_{05} f_{11} + \frac{1}{48} f_{06} \right) \\ & + \kappa^4 \left(\frac{1}{2} f_{03}^2 f_{11}^2 + \frac{5}{8} f_{03}^2 f_{12} + \frac{2}{3} f_{03} f_{04} f_{11} + \frac{7}{48} f_{03} f_{05} + \frac{1}{12} f_{04}^2 \right) \\ & + \kappa^5 \left(\frac{5}{8} f_{03}^3 f_{11} + \frac{25}{48} f_{03}^2 f_{04} \right) + \frac{5}{16} \kappa^6 f_{03}^4. \end{aligned} \tag{7}$$

This is a polynomial with 23 monomials including f_{30}, all coefficients rational. The third-genus amplitude receives contributions from the *derivatives* f_{21}, f_{22} of genus two, besides universal F_0, F_1 terms. It is not correct to compare the earlier raw YY connection cocycles directly to zero without these terms.

To verify (6)–(7) independently of graph tables, let $r = \sqrt{h}$, ξ a centered Gaussian of variance κ, and define

$$\begin{aligned} V_1 &= f_{11}\xi + f_{03}\xi^3/6, \\ V_2 &= f_{20} + f_{12}\xi^2/2 + f_{04}\xi^4/24, \\ V_3 &= f_{21}\xi + f_{13}\xi^3/6 + f_{05}\xi^5/120, \\ V_4 &= f_{30} + f_{22}\xi^2/2 + f_{14}\xi^4/24 + f_{06}\xi^6/720. \end{aligned}$$

Then exactly

$$\mathcal{F}_2^{(\beta)} = \langle V_2 + V_1^2/2 \rangle, \tag{8}$$

$$\mathcal{F}_3^{(\beta)} = \langle V_4 + V_1 V_3 + V_2^2/2 + V_1^2 V_2/2 + V_1^4/24 \rangle - \frac{1}{2}\langle V_2 + V_1^2/2 \rangle^2, \tag{9}$$

with $\langle \xi^{2n} \rangle = (2n-1)!!\kappa^n$. The subtraction is the connected-graph (logarithm) prescription.

4 Exact conifold Fourier polarization and local trivialization

The physically distinguished A-cycle at the conifold is the *vanishing* cycle, whose period is x_4. Introduce a new symplectic basis by exchanging one conjugate pair,

$$\widetilde{\Pi} = R\Pi, \qquad R = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}, \qquad (\widetilde{q}, \widetilde{p}) = (x_4, -x_2). \tag{10}$$

Direct exact matrix multiplication gives

$$R^T \Omega R = \Omega, \qquad \boxed{RM_cR^{-1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix}}. \tag{11}$$

Thus conifold monodromy becomes the *momentum* shear $\widetilde{q} \mapsto \widetilde{q}$, $\widetilde{p} \mapsto \widetilde{p} - \widetilde{q}$. In the Schrödinger convention $p = h\partial_q$ for the WKB slope, its action on wave-function components is multiplication by

$$\boxed{Z_c(\widetilde{q}; h) \longmapsto \exp\left(-\frac{\widetilde{q}^2}{2h}\right) Z_c(\widetilde{q}; h),} \tag{12}$$

up to a constant metaplectic phase and orientation convention. Accordingly, only $\mathcal{F}_0^c$ shifts by $-\widetilde{q}^2/2$, and $\mathcal{F}_g^c$ for $g \geq 1$ are unchanged by this *local monodromy*. This is an exact, nontrivial sense in which the conifold polarization makes crossing-like behavior triangular or trivial.

The operator relating LCS and this vanishing-cycle polarization is a Fourier transform in the (x_2, x_4) pair, with x_1 fixed:

$$\boxed{Z_c(\widetilde{q}; h) = \mathcal{C}(h) \int_{\mathcal{C}} dx \, \exp\left(-\frac{\widetilde{q}x}{h}\right) Z_{\mathrm{LCS}}(x; h), \qquad x = x_2.} \tag{13}$$

Its formal stationary point satisfies $\widetilde{q} = \mathcal{F}_0'(\chi)$, and the Gaussian covariance is $\kappa_c = -1/\mathcal{F}_0''(\chi)$. The genus-two and genus-three connected-graph formulas (6)–(7) therefore apply directly with $\kappa = \kappa_c$, subject to analytic continuation along the chosen path and the appropriate genus-one determinant/half-form normalization. This calculates the universal higher-genus portion of the LCS-to-conifold *polarization* change; it is not yet a numerical comparison of its projection onto the proposed finite YY bases.

The local trivialization alone does not determine the $g \geq 2$ amplitudes: every additional function of the conifold A-coordinate has the same local monodromy. Conifold gap, global continuation, and modular/GV data are additional independent inputs.

5 What happens to holomorphic ambiguities?

Suppose a genus-two ambiguity is shifted by $\delta\mathcal{F}_2 = a(x)$ while $\mathcal{F}_0, \mathcal{F}_1$ are held fixed. Independently, shift $\mathcal{F}_3$ by $b(x)$. Equations (6)–(7) give the exact *linear* transformation law

$$\boxed{\begin{aligned} \delta\mathcal{F}_2^{(\beta)} &= a(\chi), \\ \delta\mathcal{F}_3^{(\beta)} &= b(\chi) + \frac{\kappa}{2}a''(\chi) + \left(\kappa\mathcal{F}_1'(\chi) + \frac{\kappa^2}{2}\mathcal{F}_0'''(\chi)\right) a'(\chi). \end{aligned}} \tag{14}$$

In other words, the homogeneous ambiguity is *transported* by the lift; it is not killed. Even a simple nonzero $a(x) = c\,x^{-2}$ induces a genus-three term: $a' = -2cx^{-3}$, $a'' = 6cx^{-4}$. The generic transform is therefore nonzero.

For the actual quintic, a YY ambiguity polynomial $a_g(X) = \sum_{j=0}^{3g-3} c_j X^j$ is converted to the homogeneous free energy

$$\delta\mathcal{F}_g(X^0, X^1) = (X^0)^{2-2g}\,\phi_g(t), \quad t = \frac{X^1}{X^0}, \quad \phi_g(t) = \mathcal{N}_g(t)\,a_g(X(t)). \tag{15}$$

The overall period, Yukawa and gauge factors are contained in $\mathcal{N}_g$. One must *not* simply differentiate $a_g(X)$ with respect to X^0 holding X fixed. In the physical polarization the correct exact rule is

$$\partial_{X^0}^n[(X^0)^k\phi(t)] = (X^0)^{k-n}\prod_{j=0}^{n-1}\left(k - j - t\frac{d}{dt}\right)\phi(t) \qquad (X^1 \text{ held fixed}). \tag{16}$$

The genus-one amplitude furthermore carries its own scaling/half-form term. Without these conventions, *numerical* cancellation of the previously obtained P_2, P_3 cocycles has not been established.

6 An intrinsic family of completed "Polyakov-like" blocks

For a candidate insertion $a(x)$, define the metaplectically completed insertion

$$\boxed{\mathbb{B}_a^{(\beta)}(y;h) = \frac{\exp(\beta h\partial_y^2/2)[a(y)Z_0(y;h)]}{\exp(\beta h\partial_y^2/2)Z_0(y;h)}.} \tag{17}$$

The heat semigroup identity $T_{\beta_1}T_{\beta_2} = T_{\beta_1+\beta_2}$ gives a precise composition/gluing law for these objects (with the ambient wave function transformed as well). To first order in h,

$$\mathbb{B}_a^{(\beta)} = a(\chi) + h\left\{\frac{\kappa}{2}a''(\chi) + \left[\kappa\mathcal{F}_1'(\chi) + \frac{\kappa^2}{2}\mathcal{F}_0'''(\chi)\right]a'(\chi)\right\} + O(h^2). \tag{18}$$

This construction is the explicit positive result: the Gaussian lift supplies a universal, composition-compatible *completion* of each proposed block. It does *not* canonically single out a particular a in the original YY monomial space, since (17) exists for *every* insertion. Thus the stronger crossing conjecture requires an additional condition (regularity/gap at the conifold, MUM normalization, orbifold growth, integrality, and/or the modular DT/PT bootstrap) that selects a subspace of these completed blocks.

7 Further computational tests

The exact-rational implementation accompanying this note verifies the 23 genus-three terms, the genus-two formula, the shear's symplectic property, and the affine action on the higher-genus ambiguities. A complete mirror-quintic *numerical* verification would require: (i) the homogeneous prepotential and its full Hessian in Movasati's basis, (ii) the holomorphic genus-one half-form, (iii) the original YY P_2, P_3 transformed with the appropriate Yukawa/period weights, and (iv) matching all terms in (6)–(7) to their exact connection cocycles after converting from the real to the Kähler polarization. This last comparison is deliberately left open rather than inferred from the universal graph calculation.

References

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